MCP-referenced timing calibration of CMS BTL prototype bars

CMS

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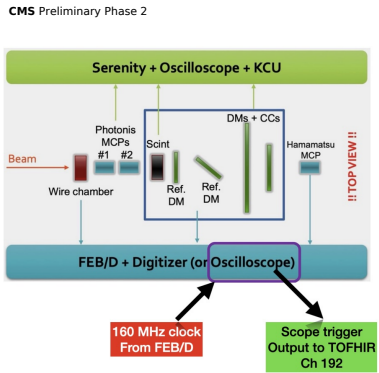
Discussion draft; not approved for public release.

Abstract

A timing-calibration procedure is studied using CMS barrel timing layer (BTL) prototype scintillator bars in the September 2025 beam test. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and a locally weighted smoothing correction are used to map the reference into the BTL time coordinate.

The dependence of the measured arrival time on pulse charge is described by quadratic functions of inverse charge for four silicon-photomultiplier channels on two reference bars. These functions provide an empirical time-walk calibration over the measured charge range. Clock-corrected profiles provide the reference for the companion dual-threshold study; archived fits are retained to document the change in calibration procedure. Timing-difference distributions illustrate the validation strategy and the need to distinguish intrinsic reference-detector resolution from residual synchronization effects. This draft documents the available results and identifies the remaining selection, uncertainty and reproducibility checks before publication.

Beam-test setup and the reference signal



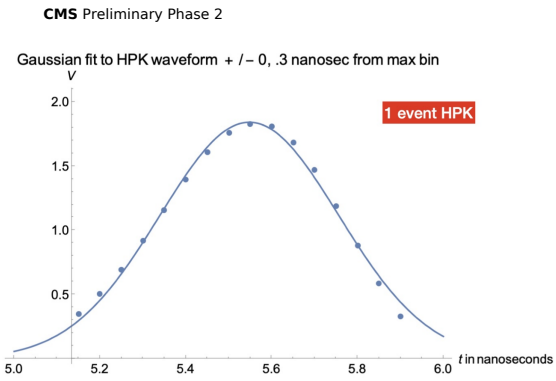
The September 2025 beam test used scintillator bars read out at both ends by silicon photomultipliers (SiPMs). A downstream MCP-PMT provides a fast reference for a restricted region of the bars.

The BTL front-end readout and oscilloscope use different time coordinates. The scope trigger reaches TOFHIR channel 192; a separate 160 MHz clock link can also be recorded by the scope.

The highlighted lower readout path is used for the reference-bar calibration.

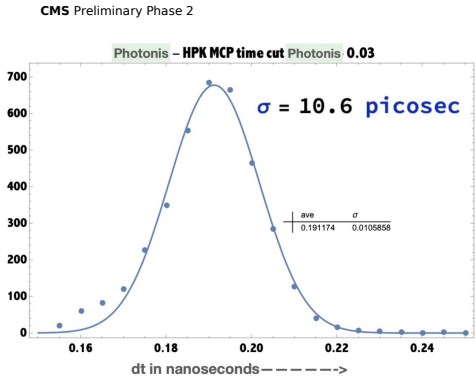
Legacy setup illustration from the supplied Note1 source. Acronyms inside the illustration will be simplified in the final redraw.

Reconstructing the MCP pulse time



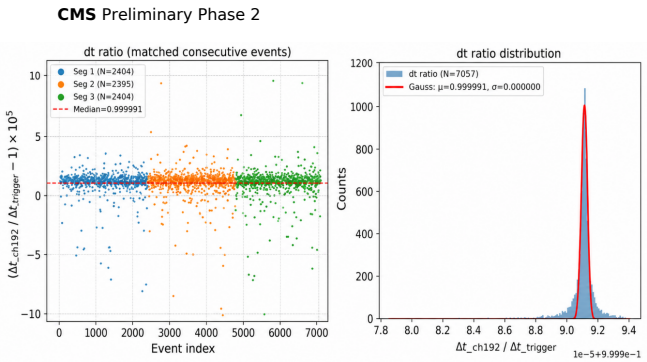
An example Hamamatsu MCP waveform is shown with its fitted pulse template. The fit estimates the pulse centroid rather than using the threshold-crossing time alone, reducing sensitivity to pulse amplitude.
The scope-trigger time identifies the event; the fitted peak supplies its relative reference time. Their difference is measured within the same oscilloscope waveform.

Intrinsic reference timing is a separate measurement



The distribution compares the fitted times of two MCP-PMTs recorded by the oscilloscope. The measured width is 10.6 ps.
Assuming independent errors and a 7.5 ps PHOTONIS contribution gives $\sigma_{\mathrm{HPK}} \simeq \sqrt{10.6^2 - 7.5^2} = 7.5$ ps.
This scope-only estimate does not include synchronization to the BTL readout.

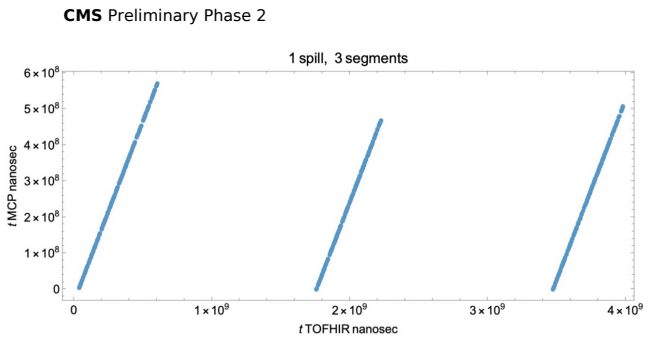
Matching the same particles between readouts



Consecutive trigger intervals measured by channel 192 are compared with those recorded by the scope. Correlated intervals identify candidate matches despite the independent clocks.
The interval-ratio distribution tests matching consistency. It is not a measurement of the picosecond resolution of the reconstructed particle time.
Only uniquely matched events enter the reference calibration.

Implementation (repository-relative):
`FEBD_Sync/simple_sync.py`
`dedup_triggers`: duplicate removal.
`match_cumulative_time`: matching and tolerance.

Clock-rate alignment within acquisition segments

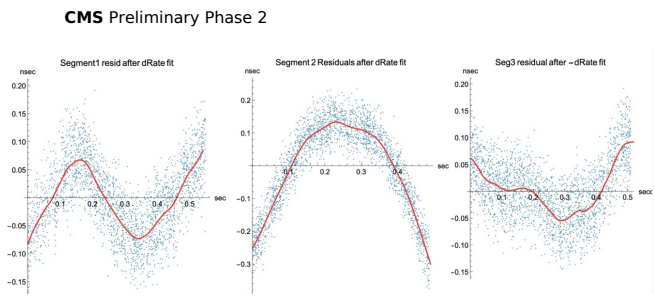


Matched scope-trigger times are compared with TOFHIR trigger times. A nearly linear relation describes the overall clock-rate difference within an acquisition segment.

An offset and a slope are needed: a slope close to one is not sufficient over a long time interval. Clock resets require separate segment coordinates.

Legacy clock-relation plot. This plot illustrates synchronization, not charge-dependent time walk.

Correcting smooth residual clock drift



After the linear clock-rate correction, the time residual still changes within each of three segments. Points are matched events; red curves show the smooth trend estimated by LOWESS, a locally weighted regression.
The fitted trend is added to the linear prediction to reconstruct the expected channel-192 trigger time. This expected time is subtracted once from the SiPM time in the reference difference.

Legacy residuals in ns versus segment time in seconds. These are pre-LOWESS residuals, not final timing resolutions. Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 657–668 and 816.

A single, explicit time convention

Let $g(u)$ map a scope-trigger timestamp u into the TOFHIR time coordinate, including the linear clock relation and the fitted smooth drift. Define the within-waveform difference

$$\delta_{\mathrm{MCP}} = t^{\mathrm{scope}}_{\mathrm{peak}} - t^{\mathrm{scope}}_{\mathrm{trigger}}.$$

The reference difference implemented in the calibration is

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t^{\mathrm{scope}}_{\mathrm{trigger}}) - \delta_{\mathrm{MCP}},$$

up to a fixed cable/geometry offset and the negligible-rate-conversion approximation for the short within-waveform interval.

The charge-dependent component of Δt_i is the time walk. An arbitrary constant does not change a shape calibration.

Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 816 and 1102. Both per-file and combined-run calculations subtract (`mcp_peak - mcp_trig`).

Reference bars and calibration selection

Upstream bar	SiPM channels	Role
7	133, 154	First reference pair
8	137, 150	Second reference pair

- Select coincident SiPM hits and a unique MCP-associated event, using channel 192 as the TOFHIR indicator.
- Require a successfully reconstructed MCP pulse and consistent peak-to-trigger timing. This is a waveform-quality cut, not a particle transit-time measurement inside the MCP.
- Use Q, the TOFHIR ADC charge proxy. It is not a deposited energy in MeV.
- Fit the profile of Δt_i versus $1/Q$ over the retained charge range.

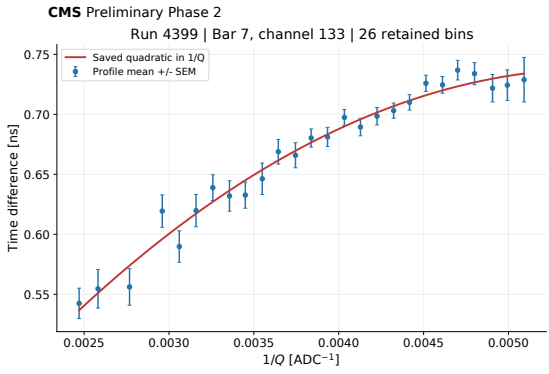
Implementation (repository-relative):

`TimeCalibration/ch192_vs_trigger_lowess.py`: coincidence and walk-sample selection.

`TimeCalibration/ch192_vs_trigger.py`: `build_mcp_map`, optional MCP cuts.

`TimeCalibration/bar_processing.py`: peak-to-trigger timing selector.

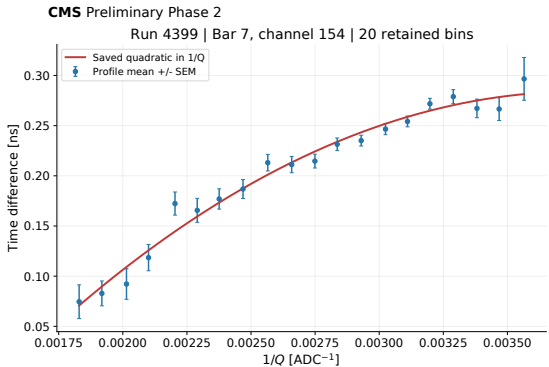
Archived reference: bar 7, channel 133



Run 4399 events are grouped in inverse-charge bins. Each point is the mean time difference; the vertical bar is its saved standard error of the mean (SEM). The curve is the saved binned quadratic $W(Q) = a + b/Q + c/Q^2$. There are 26 retained profile points. The time coordinate includes the historical constant display offset. These archived means use measured channel-192 timing. An unweighted fit to the means reproduces the saved coefficients; SEM bars are not fit weights.

Redrawn from archived profile and coefficient CSVs. No refit. Saved display convention includes a +26 ns shift.

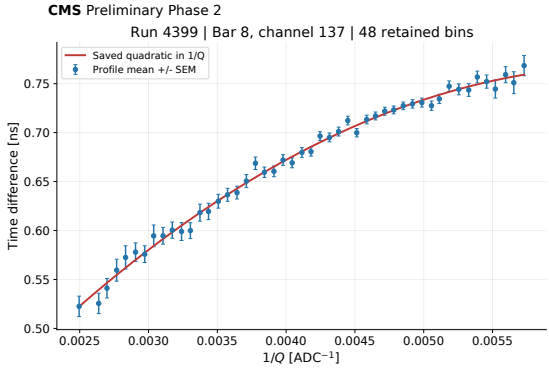
Archived reference: bar 7, channel 154



The opposite end of bar 7 is calibrated independently. Points show the mean time difference and SEM in 20 retained inverse-charge bins.
The red curve is the saved quadratic in $1/Q$, not a quadratic in Q.
Channel-dependent coefficients allow for different pulse responses at the two bar ends. The archived fit is unweighted and uses measured channel-192 timing. The retained bin counts are channel dependent.

Run 4399 legacy calibration, redrawn without refitting. Same +26 ns display convention as channel 133.

Archived reference: bar 8, channel 137

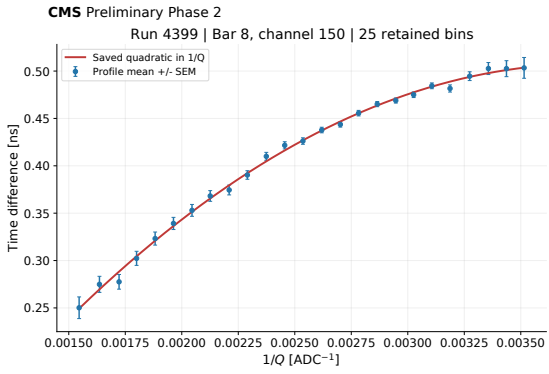


The channel-137 time difference increases as inverse charge increases: smaller signals cross a fixed threshold later.

Points are profile means with SEM; the red line is the saved binned quadratic. There are 48 retained points. The correction is applicable over the measured charge support, not automatically outside it. This unweighted, measured-channel-192 result is retained for comparison with the clock-corrected reference shown next.

Run 4399 legacy calibration. Coefficients and numerical points are included with the draft.

Archived reference: bar 8, channel 150



The opposite end of bar 8 has its own charge-dependent delay. The 25 points show the mean timing difference and SEM; the curve is the saved binned quadratic.

The four curves demonstrate the calibration procedure. Different numbers of retained points are preserved rather than imposing an undocumented common binning. This is the archived, unweighted fit using measured channel-192 timing.

Run 4399 legacy calibration. Coefficient units are ns, ns ADC and ns ADC^2 for a, b and c.

Clock-corrected reference shared with Note 2

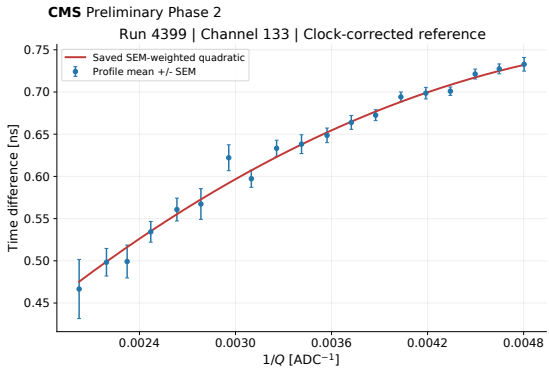
The following results use the expected trigger time $g(u)$ reconstructed from the segment-wise linear clock relation and LOWESS drift:

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - (t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}).$$

- Run 4399: exactly five TOFHIR hits, comprising the target upstream pair, downstream channels 234 and 243, and channel 192, with a unique MCP match.
- Require finite, positive charge and slope-related quality quantities; retain Q at or above the left half-maximum boundary of the spectrum peak.
- Form 20 uniform $1/Q$ bins; retain times within $5(1.4826\,\mathrm{MAD})$ of each bin's median, with at least five entries. Bars show the SEM after clipping.
- Fit $W(Q) = a + b/Q + c/Q^2$ with an objective weighted by $1/\mathrm{SEM}^2$. The next four figures use the exact saved reference coefficients used in Note 2.

Source snapshots: `2026-09-24_measured_q_overlap/inputs`. The displayed $+26$ ns constant changes neither the shape nor the saved coefficients.

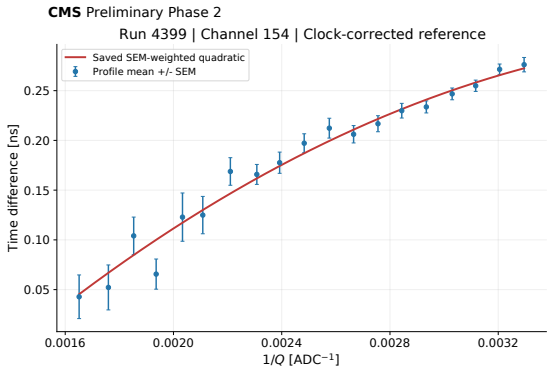
Clock-corrected reference: bar 7, channel 133



The time difference now uses the reconstructed trigger time rather than the measured channel-192 timestamp, suppressing the fitted clock drift.
Points show mean time differences with SEM in 19 retained inverse-charge bins.
The red curve is the saved SEM-weighted quadratic used in Note 2.
The curve is shown only within retained-bin support and the outer profile means. Missing bins are not extrapolated across.

Run 4399; frozen Note2 input, redrawn without refitting. A constant +26 ns shift is applied only for display.

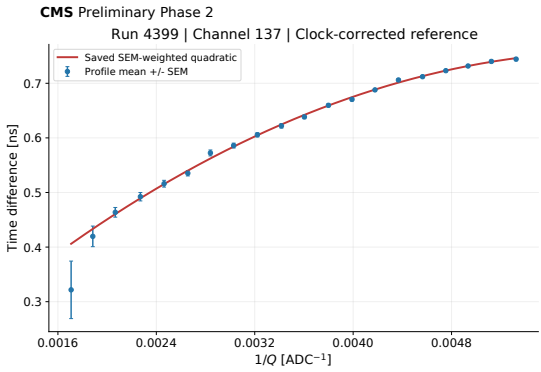
Clock-corrected reference: bar 7, channel 154



The opposite end of bar 7 is calibrated with the same clock-reconstruction and profile procedure. Its independent coefficients describe its own charge-dependent timing response. The 19 points are retained-bin means; vertical bars show SEM after timing clipping. The red curve is the saved uncertainty-weighted quadratic in $1/Q$.

Exact reference snapshot used in Note2. Coefficients and profile points are included with this note; the +26 ns display shift is separate.

Clock-corrected reference: bar 8, channel 137

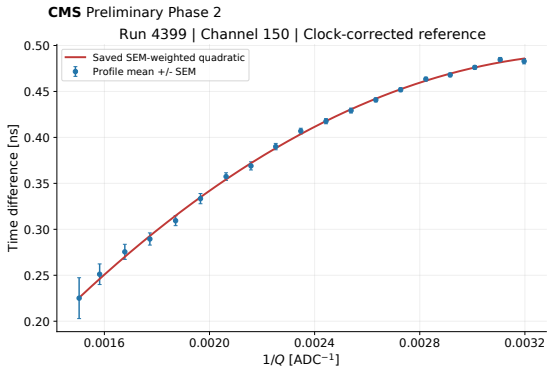


All 20 inverse-charge bins survive the profile requirements for this channel. Smaller charge gives a larger timing delay, as expected for fixed-threshold readout of pulses with different amplitudes.

Points and bars are the mean and SEM after clipping. The saved quadratic is the MCP benchmark for the channel-137 inverse-slope comparison in Note 2.

Run 4399; expected-trigger timing; no new fit or event processing. The +26 ns shift affects only the vertical display origin.

Clock-corrected reference: bar 8, channel 150



The channel-150 reference uses 19 retained profile bins and the same clock-corrected time definition as the other three channels.
Bars represent SEM, not the spread of individual event times. The red line is the saved SEM-weighted inverse-charge quadratic. These four results connect the MCP calibration in this note to the AWC shape comparison in Note 2.

Run 4399; coefficients are preserved in their original ns convention. Only the figure receives the constant +26 ns display shift.

Archived and clock-corrected shapes are close

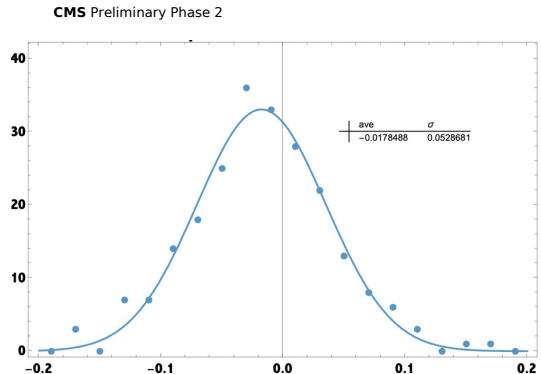
Compare both saved functions on jointly populated charge support, after removing a constant offset only. No multiplicative scaling is fitted.

Channel	Shape RMS difference [ps]	Maximum absolute difference [ps]
133	2.20	7.44
154	3.91	8.75
137	1.65	3.97
150	4.88	10.42

The walk varies by about 190–250 ps over these intervals. Rebinning the same archived events alone changes shapes by 0.14–1.41 ps RMS. The production versions also differ in trigger reconstruction, selection, clipping and fit weights.

These are differences between calibration functions, not timing resolutions or a statistical-equivalence test. Note 2 uses the clock-corrected version.

Timing validation using the two bar ends



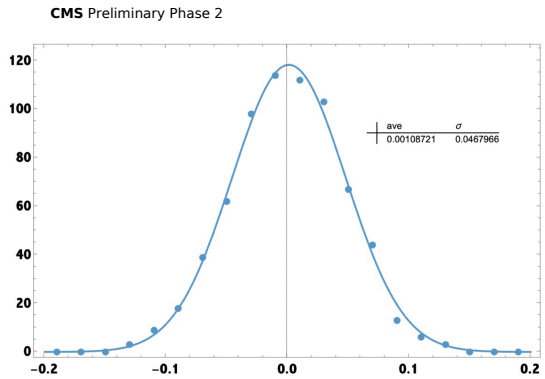
The distribution is the corrected time difference between the two SiPM ends of bar 8. The displayed Gaussian width is approximately 53 ps.

For uncorrelated end errors and controlled hit-position effects, $\sigma((t_L + t_R)/2) = \sigma(t_L - t_R)/2$, giving roughly 26 ps. Common timing jitter cancels in the end difference.

Review N1-5: Recover histogram inputs, fit range, uncertainty and position selection before treating 26 ps as a final resolution.

Legacy timing distribution. The inference is conditional; it does not measure synchronization jitter.

Timing validation against the MCP reference



The corrected bar-average time is compared with the MCP time. The legacy panel reports a width of 46.8 ps. This difference includes the bar, MCP and residual synchronization contributions, including any correlations. It must not be identified with the intrinsic MCP resolution.

Review N1-5: Establish whether 46.8 ps is a Gaussian-fit width or sample RMS, and recover its uncertainty and event selection.

Legacy bar–MCP distribution, not refitted for this draft. Do not combine unrelated selections or clock branches in quadrature.

Reference-calibration result and scope

- A trigger link permits common-event calibration across independent BTL and oscilloscope time coordinates.
- A linear clock relation plus a smooth residual correction defines the reference-time mapping.
- Four clock-corrected inverse-charge profiles and quadratic functions provide the reference used in Note 2. The archived fits remain as a comparison; numerical points, SEM and both coefficient sets accompany this draft.
- Intrinsic MCP timing, reference-bar timing and cross-system synchronization are different performance quantities.
- Charge range and operating conditions limit transfer to other samples; a constant time offset is not a walk-shape difference.

The present evidence supports the calibration workflow. Final precision claims require a frozen selection and complete statistical/systematic treatment.

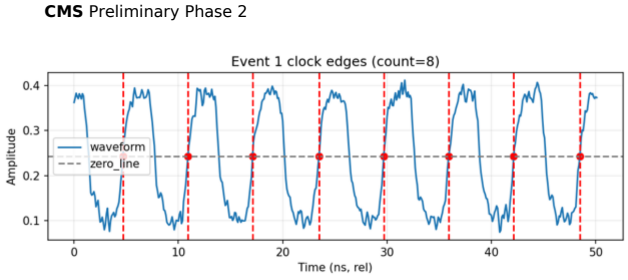
Review queue before publication

ID	Decision or missing evidence
N1-5	Reproduce bar timing widths with uncertainties and position/covariance assumptions.
N1-6	Verify citations, official template/DP metadata and final publication packaging.

Sources: supplied `Note1.pdf`; candidate editable manuscript `CMS-Note/cmsnote`; archived calibration CSVs in `Q_Walk/reference_quadratic_t0`.

The figure manifest records the exact copied inputs and checksums. This review slide is removed from a public release.

Backup: clock-phase reconstruction



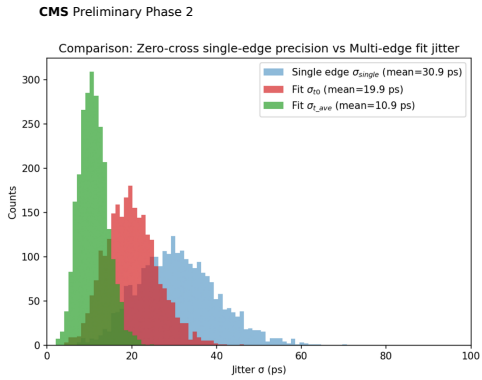
The reference clock is sampled by the oscilloscope. Several rising-edge crossings can be reconstructed within a waveform.

Fitting edge time against edge index allows the clock phase to be estimated using multiple cycles. Evaluating the fit near the mean edge index avoids unnecessary extrapolation.

This is an alternative synchronization diagnostic, distinct from the trigger-link branch used for the legacy reference curves.

Legacy illustration. Correlated edge errors prevent assuming an automatic $1/\sqrt{N}$ precision improvement.

Backup: comparing clock estimators



The recorded MCP time is compared with three clock-phase estimators: a single edge, a regression intercept and the fitted time at the mean edge index. The distributions illustrate estimator dependence. Their widths contain the contributions present in this specific measurement and are not automatically an absolute clock-accuracy measurement.

Review N1-6: Recover source arrays and confirm common selections and width definitions before using numerical improvements.

Legacy clock-estimator diagnostic, retained as a discussion backup.